

Proofs should be written neatly and concisely. You may quote theorems from lecture, but any other results need to be proven. Make sure to reference all sources of collaboration.

- (1) Prove that $\mathbb{R}P^1 \simeq S^1$. (One possible way to do this is to construct an explicit map and prove it is a homeomorphism.)
- (2) (a) Give n distinct homeomorphisms producing copies of $\mathbb{C}P^{n-1}$ in $\mathbb{C}P^n$.
 (b) Let $q : S^{2n+1} \rightarrow \mathbb{C}P^n$ be the quotient map and $\phi : B^{2n} \rightarrow \mathbb{C}P^n$ be a map from the unit ball to complex projective space defined by

$$\phi(z) = q(z, \sqrt{1 - |z|^2}).$$

Prove that $\phi(S^{2n-1}) = \mathbb{C}P^{n-1} \subset \mathbb{C}P^n$.

- (c) Prove that $\mathbb{C}P^{n+1}/\mathbb{C}P^n \simeq S^{2n+2}$.
- (3) Prove that $S^n \wedge S^m \simeq S^{n+m}$ for any n and m .
- (4) Let X be a locally compact Hausdorff space. Given a point $x \in X$ and a neighborhood U of x , show that there is an open set V such that $x \in V \subset \bar{V} \subset U$ and $\bar{V}$ is compact.
- (5) Consider the smash product $X \wedge Y$ of two pointed spaces X and Y . Viewing this as a product, what is the unit E , i.e. for which topological space E is $X \wedge E \simeq X$? Now answer the same question for the wedge sum $X \vee Y$.
- (6) Consider $X = Y = \mathbb{Q}$ and $Z = \mathbb{N}$ with the subspace topology and with basepoint 0. Then

$$(X \wedge Y) \wedge Z \not\simeq X \wedge (Y \wedge Z).$$

This shows that for pointed topological spaces, the smash product is not associative.

Your task (should you accept it, and you really should!) is to either

- (a) prove this or
- (b) find a proof and write a brief sketch of it.

Remark: I know of only two "proofs" of this claim, first made in 1958 by Puppe. One of the "proofs" is at best a sketch, I do not think it is correct. The second was published in 2006 in a book by May-Sigurdsson (the actual counterexample being included in an appendix and having been worked out by Kathleen Lewis). This might serve as a hint as to which of the above you should try to do.